

MIRUL KHANAL

FULL-STACK ENGINEER

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 Kathmandu

PROFILE

With over 5 years of experience as a full-stack engineer, I have contributed to the architecture and development of scalable SaaS platforms, high-performance mobile applications, and AI-driven systems including: multi-tenant ecosystems, OCR/NLP pipelines, and big data optimization using tools like PySpark and Elasticsearch. I’ve worked across the full stack, from React and Next.js frontends to FastAPI and Node.js backends, while also implementing security and compliance standards such as HIPAA and MFA. I take a practical, business-focused approach to engineering, and I’m confident in my ability to quickly master new technologies whether it’s a new framework, a data processing tool, or an AI model and apply them effectively within short timeframes to deliver production-ready solutions.

CORE SKILLS

LANGUAGES JavaScript (ES6+), TypeScript, Python, PHP, Node.js, Express, FastAPI, Java (Intermediate – Android, Spring/Spring Boot), Bash, Go (familiarity – personal projects)	FRONTEND React, Next.js, Vue.js, React Native, Tailwind CSS
BACKEND Multi-tenant SaaS, REST APIs, Auth (OTP, MFA), Prisma, GraphQL, Socket.IO, NestJS	AI & DATA PaddleOCR, BERT, ONNX, Elasticsearch, PySpark, LookerML, Amplitude
INFRASTRUCTURE PostgreSQL, MongoDB, AWS S3, CI/CD, GitHub CI, Docker & Docker Compose, Linux, HIPAA Compliance, Agile (JIRA)	BAAS & TOOLS Firebase (High), Supabase (High), PocketBase (Mid)

EXPERIENCE

○ Milo Logic

2025 – 2026

Full Stack Developer

Kathmandu

MUs (Multi-Tenant SaaS): Pioneered core backend architecture for a multi-tenant ecosystem with dynamic tagging and global filtering, improving data discoverability.

Infrastructure Innovation: Orchestrated custom domain routing ("Bring Your Own Domain") and cross-tenant membership with granular RBAC.

Cyberport OCR Pipeline: Built end-to-end AI pipeline using PaddleOCR and BERT for high-accuracy structured key-value extraction from varied receipt formats.

Mobile Optimization: Deployed lightweight ONNX-based OCR models directly on mobile devices for offline processing and reduced server latency.

Security & UI: Fortified platform security with MFA and OTP authentication; developed BAUM frontend and CP OCR Admin Panel using Next.js.

○ Deerhold Technologies

2023 – 2025

Software Engineer

Kathmandu

TARA (Big Data Optimization): Re-engineered Elasticsearch ingestion pipelines using PySpark, achieving performance gains through memory optimization, sharding, and cluster tuning.

HIGI Health (Frontend & Analytics): Led frontend development for large-scale KIOSK systems using React; integrated Amplitude for event-based tracking and LookerML for reporting.

Valenz Health (Security & Modern Web): Implemented HIPAA-compliant security practices; developed high-performance modules using FastAPI and Vue.js.

Enterprise Workflow: Streamlined delivery using JIRA and GitHub CI/CD with automated PR/Code reviews and robust health data anonymity.

Remote Systems Management: Established secure protocols for remote KIOSK access to update production code in isolated environments.

Technical Leadership: Steered digital strategy and system architecture for diverse client projects, bridging business objectives with technical execution.

Process Excellence: Transformed internal workflows with rigorous CI/CD pipelines and automated testing, boosting quality and delivery speed.

○ TechTemple IT Solutions

2021 – 2022

Full Stack Developer

Butwal

Product Delivery: Steered the digital strategy and system architecture for diverse client projects, bridging the gap between business objectives and technical execution.

Project/Process Management Process Excellence: Transformed internal delivery workflows by implementing rigorous CI/CD pipelines and automated testing, resulting in higher code quality and faster ship cycles.

○ TechTemple IT Solutions

2020 – 2021

Chief Digital Officer

Butwal

Technical Leadership: Delivered production-ready features for GSAtHome (MERN), MedicosPDF (React Native), and Waymandu (Laravel + React).

Agile Development: Played a pivotal role in agile sprints, consistently shipping scalable features across web and mobile platforms.

○ Macquarie Group

2017 – 2018

Intern (Through La Trobe)

Sydney, Australia

ML Research: Contributed to fraud detection initiatives by developing machine learning models using TensorFlow and Keras, gaining deep insights into predictive analytics.

EDUCATION

BSc Computer Science

NAMI College, Kathmandu

A Levels

Xavier International College, Kathmandu

Diploma in Computer Programming

La Trobe University, Sydney, Australia

Awarded Scholarship from NAVITAS for excellent performance

REFERENCES

Mark Fox

Academic Coordinator
La Trobe University

Shashank Adhikari

CEO
TechTemple IT Solutions

Sachin Karanjit

Director of Operations
Deerhold Technologies

Bipin Shakya

CEO
Milo Logic